8D report

Customer plant:	Mahle Sempeter	**Supplier:**	AVO Carbon Germany	**Date of claim:**	25.08.2025
Customer contact:	Saza Babic	**AVO contact:**	Gunther Luschtinetz		
Customer reference:	200123836	**AVO reference:**	250809		

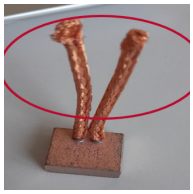
QUICK RESPONSE (D1-D3)
This document must be issued within 24 hours after claim notification

D 1. Description of the problem

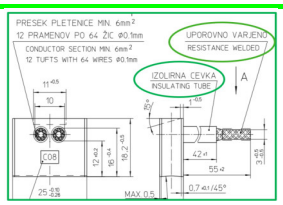
1.a What is the problem? Use 5W+2H to describe what is the current situation.

Problem description	Missing insulating tube Stranded wire is not resistance welded	What happened	Missing insulating tube Stranded wire is not resistance welded
		Why is it a problem	Parts out of spec.
Incident type	Customer claim		
Part number AVO	1026515	When did it happen	26.08.2025
Part number Customer	16770253	Who detected it	Incoming inspection
Batch number	0001530436	Where was it detected	Incoming inspection
Recurrence	Non-recurrent	How was it detected	Visually
Defect S/R	No	How many bad parts	2000
Standard process used	Yes		
When was the part produced?	Pressing machine => not relevant - welding machine in POI Grinding machine => not relevant, because no grinding error		

nOK part picture/ sketch



OK part picture/ sketch



1.b Problem solving team

Name	Position	Location	E-mail
Gunther Luschtinetz	Quality Management	AVO Germany	joachim.gehl@avocarbon.com
Gunther Luschtinetz	Quality Management	AVO Germany	gunther.luschtinetz@avocarbon.com
Martin Röhr	Production	AVO Germany	
Mustafa Yorulmaz	Production	AVO Germany	
Khaled Amri	Logistic	AVO Germany	

1.c Tools to use

Yes	**Ishikawa**	Yes	**5why's**		**LLC**		**Audit**		**Other**

1.d First analyse of problem / potential root causes

Potential root cause(s): Work plan not correct

If you use our expressnumber, please use only economic express.

Samples of nok parts required in plant for analyse: Yes

Express shipping number account: FedEx 186141300

D 2. Risk on similar products and processes

Can this defect happen on similar parts?		Comments (compulsory, if yes)
Other family product	No	This is the only brush type where the 2 wires are welded together.
Other line	No	
Other tools/work station	No	
Other	No	

D 3. Containment

Sorting criterias (add additional sheets if needed)

Location in the supply chain	N° of involved parts	N° parts sorted	N° nok parts	Ppm	Date	Responsible
Parts at customer	2.000	---	2.000	---	26.08.2025	Saza Babic
Parts in transport	---	---	---	---	---	---
Parts at AVO Germany stock	0	0	0	0	26.08.2025	Gunther Luschtinetz
Parts at AVO production	---	---	---	---	---	---

First contained shipment information

Immediate actions to prevent production and delivery of nok parts ?

Containment actions	Responsible	Date planned	Date closed
Register the complaint + Quality alert	Gunther Luschtinetz	26.08.2025	26.08.2025
Check the production-, quality- and maintenance data	Gunther Luschtinetz	26.08.2025	26.08.2025
Information of all department leader during shop floor meeting QRQC	Joachim Gehl	26.08.2025	26.08.2025
Information of the sub-supplier AVO Poitiers	Joachim Gehl	26.08.2025	26.08.2025

8D report

PLAN - DO (D4-D6)

D 4. Root cause of occurrence

Work out the root causes using Ishikawa and 5 why's

Root cause of occurrence :

The work plan had not been set up correctly. During its creation, the welding process carried out at our plant in Poitier was not included. After the final processing step, the parts were stored as finished goods in accordance with the work plan. Unfortunately, these parts were shipped to Mahle.
A work instruction for creating work plans is not clearly defined and does not include a double check.

There is no work instruction that clearly defines the process for creating work plans. As a result, there is also no double check in place to prevent failures during work plan creation.

How was the defect reproduced ?

Due to the non-technical nature of the defect, it was not necessary to reproduce the cause.

D 5. Root cause of non detection

Work out the root causes using Ishikawa and 5 why's

Root cause of non detection :

Shipping staff were unaware that unfinished parts were going directly to customers. The risk of missing part in the work

D 6. Corrective actions and non detection

Corrective actions	Responsible	Date planned	Date closed
Correct the failure in the work plans. Add the extended workbench Poitiers to avoid incorrect deliveries.	Z. Savicic	25.08.2025	23.09.2025
Create a work instruction to create work plans that include a double check	H. Semo	01.10.2025	06.10.2025
Training of the relevant employees	H. Semo	07.10.2025	
Update the FMEA	G. Luschtinetz	29.09.2025	29.09.2025

CHECK - ACT (D7-D8)

D 7. Effectiveness of action plan

Actions to check effectiveness (to plan with part D6)	Responsible	Date planned	Date closed
Check the change in the work plan	QM	23.09.2025	23.09.2025
Review the new documents	D. Beil	03.10.2025	06.10.2025
Review the updated FMEA	M. Ott	29.09.2025	29.09.2025

D 8. Lessons learned

8.1 Do following documents require an update ?

	Responsible	Date planned	Date closed
P-FMEA	G. Luschtinetz	29.09.2025	29.09.2025
Control Plan			
Procedure			
Instructions	H. Semo	01.10.2025	07.10.2025
Other	Z. Savicic	25.08.25	23.09.25
Other			

8.2 Generalization of countermeasures

Double check the other work plans of correctiveness after the training.

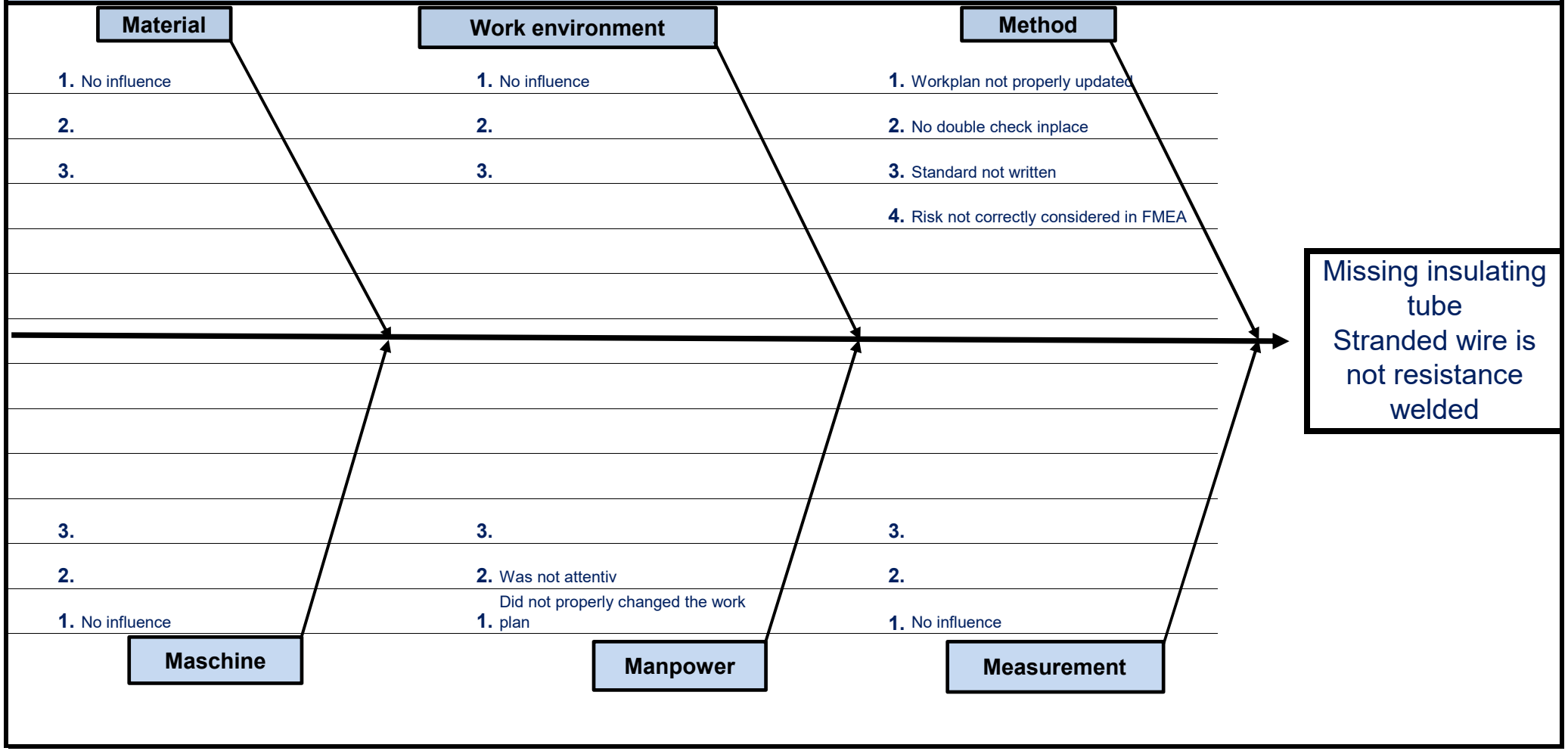
Generally, if one of the tools listed above is used. This is carefully checked and updated.

8.3 Internal audit (if required)

Date Results

Comments

Ishikawa


Material
1. No influence
2.
3.
Work environment
1. No influence
2.
3.
Method
1. Workplan not properly updated
2. No double check inplace
3. Standard not written
4. Risk not correctly considered in FMEA
Missing insulating tube
Stranded wire is not resistance welded
3.
2.
1. No influence
Maschine
3.
2. Was not attentiv
1. Did not properly changed the work plan
Manpower
3.
2.
1. No influence
Measurement

5 Why's

Problem description : Missing insulating tubeStranded wire is not resistance welded

OCCURENCE

Problem	N°	1st WHY?	1st ANSWER	2nd WHY?	2nd ANSWER	3rd WHY?	3rd ANSWER	4th WHY?	4th ANSWER	5th WHY?	5th ANSWER
	1	Why were the parts not welded?	Because the welding step was not done.	Why was the welding step not done?	Because the parts were stocked as finished parts.	Why were the parts stocked as finished parts?	Because the work plan was not correct.	Why was the work plan not correct?	Because the employee did a mistake during creation of the work plan.	Why did the employee a mistake during the creation of the work plan?	Because a double check of this important document was not planned.
		6st WHY?	**6th ANSWER**	**7th WHY?**	**7th ANSWER**						
		Why was no double check of this important document planned.	Because there is no work instruction to create work plans.	Why is there no work instruction to create work plans?	Because the risk was not considered in the FMEA.						

NON-DETECTION

Problem	N°	1st WHY?	1st ANSWER	2nd WHY?	2nd ANSWER	3rd WHY?	3rd ANSWER	4th WHY?	4th ANSWER	5th WHY ?	5th ANSWER
	1	Why were not welded parts delivered?	Because the parts were stocked as finished good parts.	Why were the parts stocked as finished good parts?	Because it was not possible to detect the parts.	Why was it not possible to detect the parts?	Because the employee followed the work plan.				

Work plan old

```
                                                    Linie/Zelle. . *
Aktionscode. . . . I
Teilenummer. . . . 1026515                          ML 16.770.253    4D52099
Stapelmenge. . . .                  CT              Zeichngsnr
Stand am . . . . . *                                Sprung zu Arbgang

      Arb       Arbgg                                  . .Laufzeit .    Rüstzt
O   platz       SeqNr     Beschreibung               Masch    Pers.     Dauer
    8005       5,00 Material bereitstellen            0,00     0,00     0,000
    1721      10,00 MISCH.HOL.PRESSEN                 0,00     7,94     4,000
    0703      20,00 GLUEHEN                           0,00     1,44     1,500
    0703      25,00 QS nach Gluehen                   0,00     0,00     0,000
    5202      50,00 ABWIEGEN UND VERPACKEN            0,00     0,36     0,000
    8995     995,00 Fertigmelden                      0,00     0,00     0,000
```

Workplan new

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 3003                Arbeitsplan anzeigen          NDL/Werk . . .            01
                                                   Artkel - Prüf
                                                   Arbplanart . . M
                                                   Linie/Zelle. . *
Aktionscode. . . . I                               verl. Werkbank Poitiers
Teilenummer. . . . 1026515                         ML 16.770.253    4D52099
Stapelmenge. . . .                  CT             Zeichngsnr
Stand am . . . . . *                               Sprung zu Arbgang

      Arb       Arbgg                                  . .Laufzeit .    Rüstzt   C
O   platz       SeqNr     Beschreibung               Masch    Pers.     Dauer   P
    8005       5,00 Material bereitstellen            0,00     0,00     0,000
    1721      10,00 MISCH.HOL.PRESSEN                 0,00     7,94     4,000   K
    0703      20,00 GLUEHEN                           0,00     1,44     1,500
    0703      25,00 QS nach Gluehen                   0,00     0,00     0,000
    5202      50,00 ABWIEGEN UND VERPACKEN            0,00     0,36     0,000
    8995     995,00 Fertigmelden                      0,00     0,00     0,000
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D 6. Corrective actions and non detection

Function | Failure type | impact | Root cause | prevention | Detection | Mesasure

Komponente Prozess	Funktion Zweck	Fehlerart	Besondere Merkmale	Fehler-Auswirkung	Fehler-Ursache	Fehler-Vermeidung	Fehler-Entdeckung	A	B	E	AxBxE=RPZ	Maßnahme	Verant-wortl.	Termin	A	B	E	AxBxE=RZ
	Fehlende Bearbeitung Reklamation Mahle 250701/ 250809	Teile ohne Bearbeitung		Teile können nicht verbaut werden	Arbeitsplan im JDE unvollständig	Arbeitsplan aktualisieren	Kann nur beim Kunden bemerkt werden	2	8	8	128	Arbeitsplan aktualisieren	Z. Savicic		4	4	4	64

Root cause | Parts without machining | Parts cannot be used | Update incomlete | Update | Can only be detected at customer site | Update work plan